ReValo™
Developed by Pranesh Sai Krishnan
2026

# ReValo

## Intelligent Waste-to-Wealth Resource Management & Urban Mining Network
## TEAM MIQ • Prototype: AI-Powered Robotic Waste Segregation

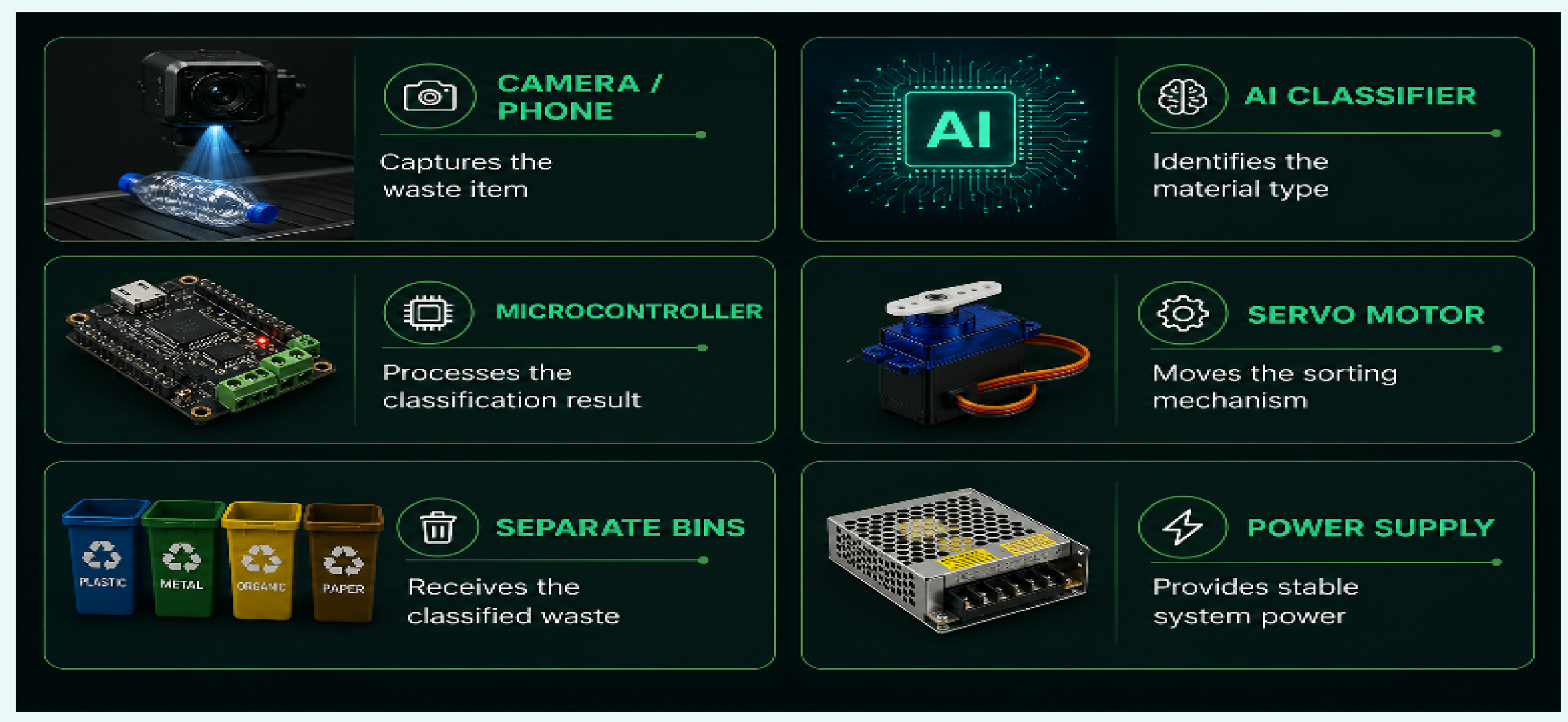
RECOVER • REVALUE • REINTEGRATE
From waste management to resource intelligence

### INTRODUCTION

Traditional mixed-waste handling causes contamination, material loss and landfill dependency. ReValo reframes the challenge as resource recovery: a camera-based AI recognition system identifies waste, while a robotic sorting mechanism routes each item into a recoverable stream. The prototype is the sensing and sorting layer of a larger intelligent resource-recovery network.

### KEY COMPONENTS

- **Camera / Phone — captures the live visual feed.**
- **AI Classifier — identifies the material category.**
- **Microcontroller — processes classification and triggers action.**
- **Servo Motor — moves the sorting mechanism.**
- **Recovery Bins — receive classified material streams.**
- **Power & Data Supply — supports operation and recovery-data logging.**




### SCOPE OF THE PROJECT

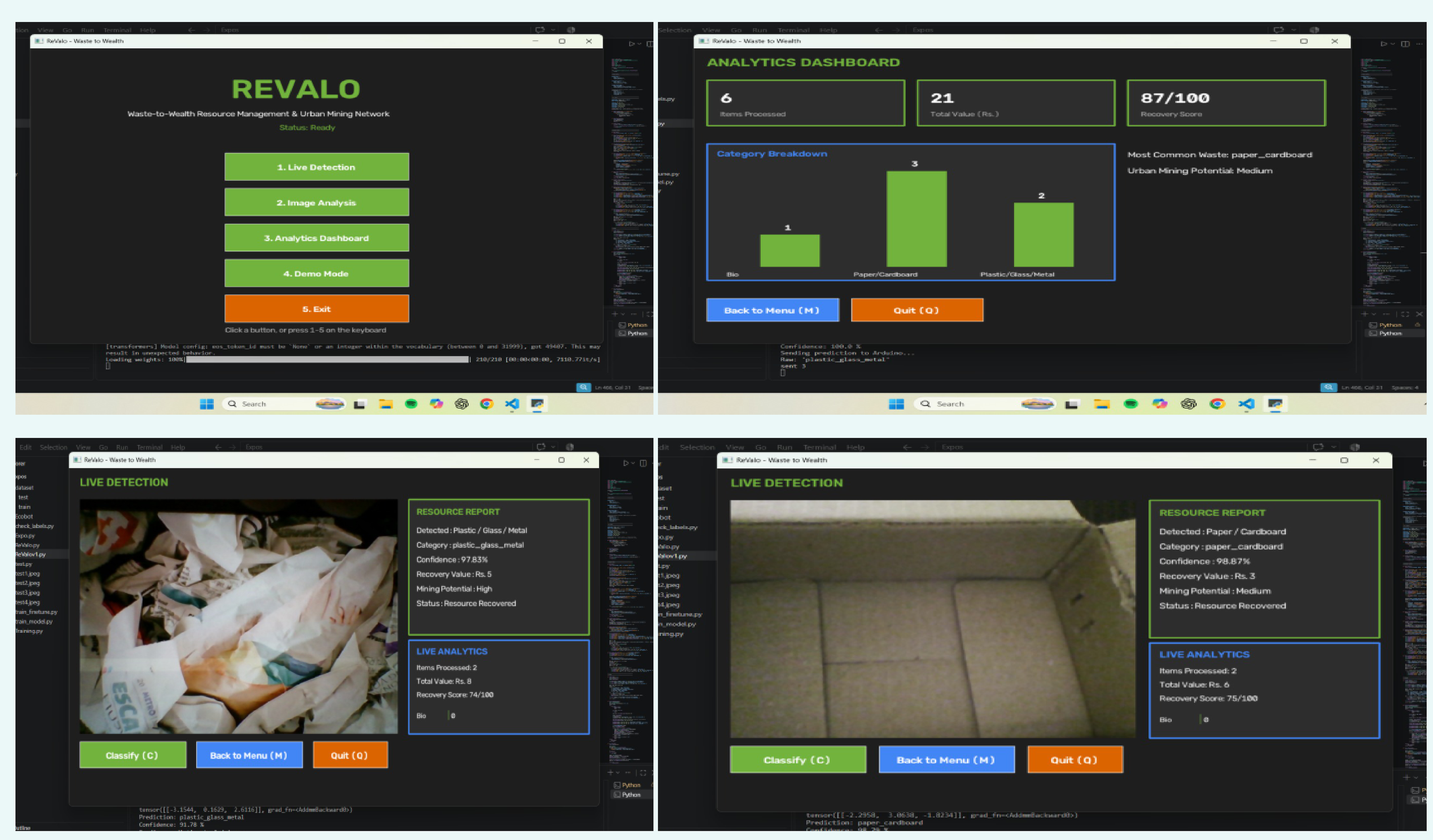
- 1. CAPTURE — camera observes the waste item on the conveyor.
- 2. IDENTIFY — AI classifier predicts the material category.
- 3. DECIDE — controller selects the correct recovery bin.
- 4. SORT — robotic mechanism directs the item for recovery.

### WORKING PRINCIPLE

- Recover valuable materials before they reach landfill.
- Use AI vision and Machine Learning Classification to identify material categories in real time.
- Automate physical sorting with a robotic mechanism.
- Create cleaner, traceable recovery streams for reuse and recycling.
- Build a scalable foundation for resource intelligence and urban mining.

### PROTOTYPE VISUAL

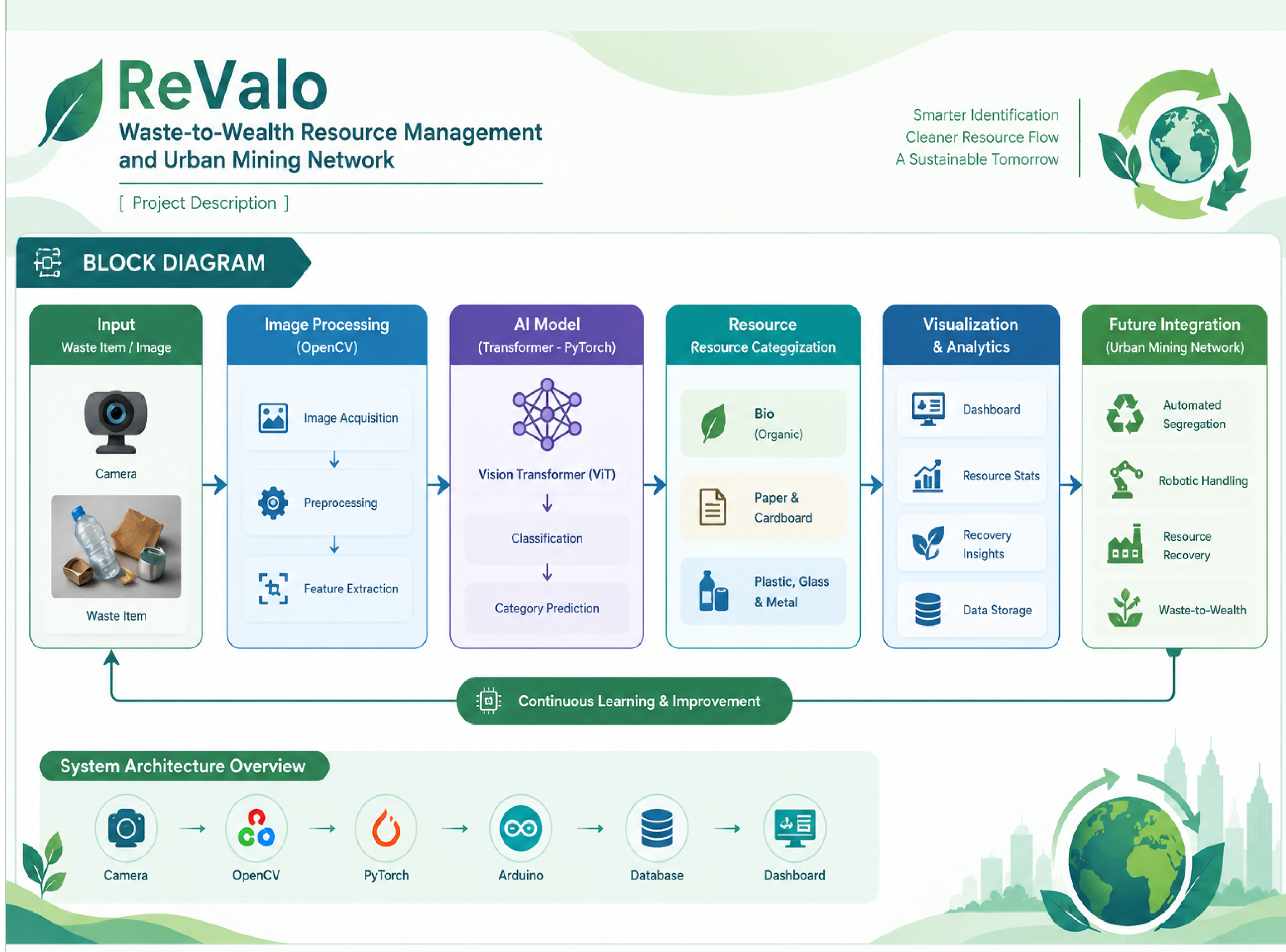
- ILLUSTRATIVE IMPACT (PROJECTED, NOT MEASURED)
- Resource recovery: 35% → 70%
- Contamination rate: 40% → 15%
- Recycling efficiency: 45% → 80%
- Resource value retention: 30% → 75%
- 100 kg mixed waste example: ~30 kg plastic, ~10 kg metal, ~20 kg paper, ~15 kg cardboard.




Prototype architecture: Camera/Phone • AI Classifier • Microcontroller • Servo Motor • Recovery Bins • Power Supply

### BLOCK DIAGRAM

- Resource → Product → Use → Recovery → Reuse → Economic Value
- Plastic: bottles, wrappers, containers
- Metal: cans and metal objects
- Degradable: food / organic waste
- Non-recoverable: other unsuitable waste
- Clean, contamination-free streams improve downstream recovery and resale potential.




### APPLICATIONS

- Campus waste segregation and sustainability monitoring
- Municipal waste-recovery and landfill-burden reduction
- Cleaner material streams for recycling and industry partners
- Resource tracking, analytics and future circular-supply-chain integration
- Future deployment: campuses, streets, buildings and industries